



Xi'an Jiaotong-Liverpool University

西交利物浦大學

WebApp Deployment

Chengtao Ji

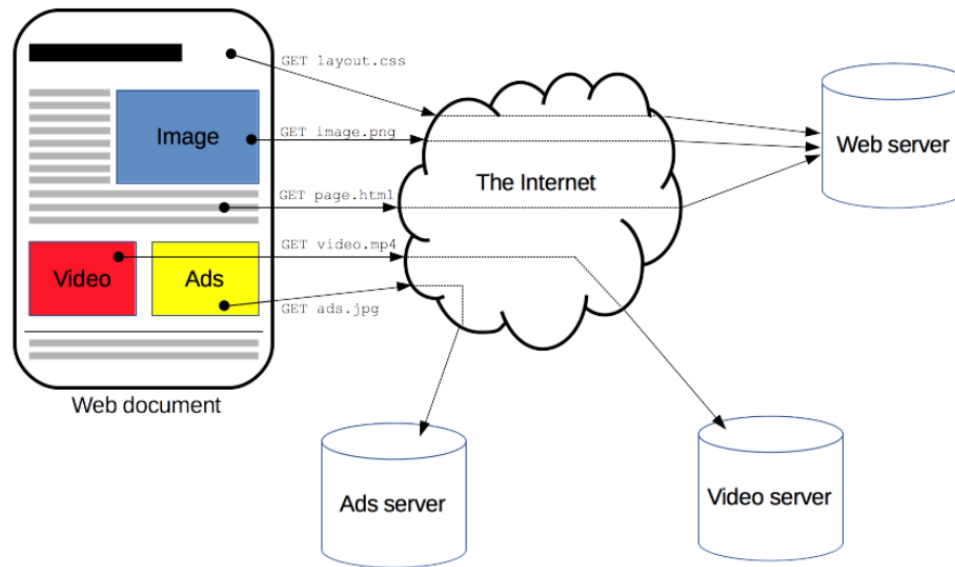
Computing Department

Web Server

A web server is software and hardware that uses HTTP (Hypertext Transfer Protocol) and other protocols to respond to client requests made over the World Wide Web. The main job of a web server is to display website content through storing, processing and delivering webpages to users. Besides HTTP, web servers also support SMTP (Simple Mail Transfer Protocol) and FTP (File Transfer Protocol), used for email, file transfer and storage.

Web server hardware is connected to the internet and allows data to be exchanged with other connected devices, while web server software controls how a user accesses hosted files. The web server process is an example of the client/server model. All computers that host websites must have web server software.

(<https://whatis.techtarget.com/definition/Web-server>)



<https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview>

ECS instance

Elastic Computer Service (ECS) is essentially a server, you will get complete control over what you can do. It offers all the features of physical servers without the overhead of managing the physical hardware. With ECS, you have the freedom of attaching the disks for extra storage, using backup images/new images, defining security groups & snapshots. These are available only with ECS Instance.

The image displays three sequential screenshots of the Alibaba Cloud RAM User Login interface.

Screenshot 1: RAM User Login
The interface shows a message about Multi-Factor Authentication (MFA) being enabled. Below the message, the "Username" field is populated with "<user>@<domain> or <user>@<alias> user name". A red arrow points to the "user name" text. The "Next" button is visible at the bottom.

Screenshot 2: RAM User Login
The interface shows the "Password" field populated with "password". A red arrow points to the "password" text. The "Log On" button is visible at the bottom.

Screenshot 3: Choose MFA Device
The interface shows the "Choose MFA Device" section. A red arrow points to the "Skip this" link in the top right corner. The "SMS Verification" option is selected, and the "Virtual MFA Device" option is also visible.

<https://signin.aliyun.com/login.htm#/main>

Cpt202@1553634882117581.onaliyun.com

ECS instance

Elastic Computer Service (ECS) is essentially a server, you will get complete control over what you can do. It offers all the features of physical servers without the overhead of managing the physical hardware. With ECS, you have the freedom of attaching the disks for extra storage, using backup images/new images, defining security groups & snapshots. These are available only with ECS Instance.

The image displays three sequential screenshots of the Alibaba Cloud RAM User Login interface.

Screenshot 1: Username Entry
The page title is "RAM User Login". A message states: "In order to better protect the security of your account and assets, starting from August 20, 2024, multi-factor authentication (MFA) will be enabled for all users when logging in. For details, see Announcement. Starting from September 20, 2024, idle RAM users who have not logged in for more than 2 years will be automatically disabled in batches. For details, see Announcement". The "Username" field is labeled with an asterisk and contains the text "<user>@<domain> or <user>@<alias> user name". Below the field, an example is provided: "Example: username@company.onaliyun.com or username@company-alias.". A blue "Next" button is at the bottom. A link "Login with Alibaba Cloud Account" is at the bottom left. The footer says "Logging in and using RAM with acceptance of the Terms of Service".

Screenshot 2: Password Entry
The page title is "RAM User Login". The same MFA announcement is shown. Below the header, it says "Hi Cpt202". The "Password" field is labeled with an asterisk and contains the text "password". Below the field, a message states: "If you forget your password, use your Alibaba Cloud account to log on to the RAM console and reset the password." A blue "Log On" button is at the bottom. A link "Login with Alibaba Cloud Account" is at the bottom left. The footer says "Login using passkeys is quicker and more secure".

Screenshot 3: MFA Device Selection
The page title is "Choose MFA Device". It states: "You currently do not have an available multi-factor authentication device. We strongly recommend that you bind one to prevent identity theft and protect the security of your account assets." Below this, it says: "Click Options and select a different MFA device to bind." A yellow box contains a message: "In order to better protect the security of your account and assets, starting from August 26th, 2024 gradually go online by account UID, RAM users who are granted AdministratorAccess system permission must perform MFA verification when logging in. For details, please see the announcement." On the right, there are three options: "SMS Verification" (with a "Skip this" link and a red arrow pointing to it), "Virtual MFA Device" (with a link "What is VMFA?"), and "Passkey". Each option has a right arrow.

<https://signin.aliyun.com/login.htm#/main>

Cpt202@1553634882117581.onaliyun.com

ECS instance

Recent Visits

Product Catalog View Resource List View

All China (Hangzhou) 3

Select resource tags Select a resource group Select the product category Sort by Recommendation

Computing

Elastic Compute Service > 1, Click Instance

1 Instance 1 Disk 1 Elastic network card

Governance Maturity (Best Practices)

Instance

Instance 1 Available Resources - By Region China (Hangzhou) 1

Instance ID/Name	Region	Resource Group	Tag	Act...
i-bp1b0vz3bpkmvuu7d19 Cpt202	China (Hangzhou)	Cpt202		View

Previous Next Items per Page 10

Elastic Compute Service / Instance / i-bp1b0vz3bpkmvuu7d19

← Cpt202 Running Expiring

Instance Details Monitoring Security Groups Block Storage Snapshot-consistent Groups Snapshots ENIs Scheduled and Automated Tasks Operation Records Health Checks Events

Basic Information

Instance ID i-bp1b0vz3bpkmvuu7d19 Connect Reset Password

Instance Name Cpt202

Zone Hangzhou Zone H

Health Status Normal

Configuration Information

Instance Type ecs.e-c1m2.large Change Purchase Same Specifications

CPU and Memory 4 Cores (vCPU) 8 GiB Change CPU Options Change Configurations

Public IP Address 47.96.158.212 Convert to EIP

Operating System Windows Server 2025 DataCenter Edition 64bit English Edition Change Image

Alerts for Critical Events (0) View Details

Installed Extensions (0) Install Extension

Security Protection

Security Center System Vulnerability Weak Password Alert

Anti-DDoS Origin Basic (Free)

Elastic Compute Service / Instance / i-bp1b0vz3bpkmvuu7d19

← Cpt202 Running Expiring

Instance Details Monitoring Security Groups Block Storage Snapshot-consistent Groups Snapshots ENIs Scheduled and Automated Tasks Operation Records Hr

Basic Information

Instance ID i-bp1b0vz3bpkmvuu7d19 Connect Reset Password

Instance Name Cpt202

Zone Hangzhou Zone H

Health Status Normal

Configuration Information

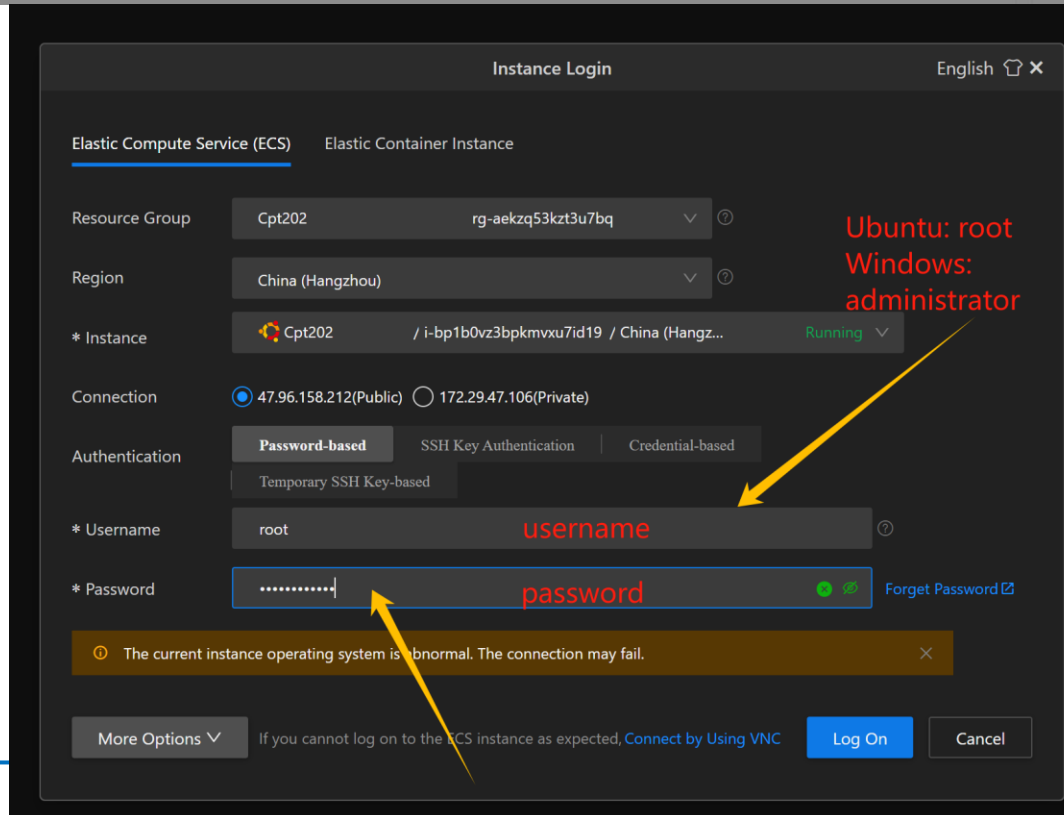
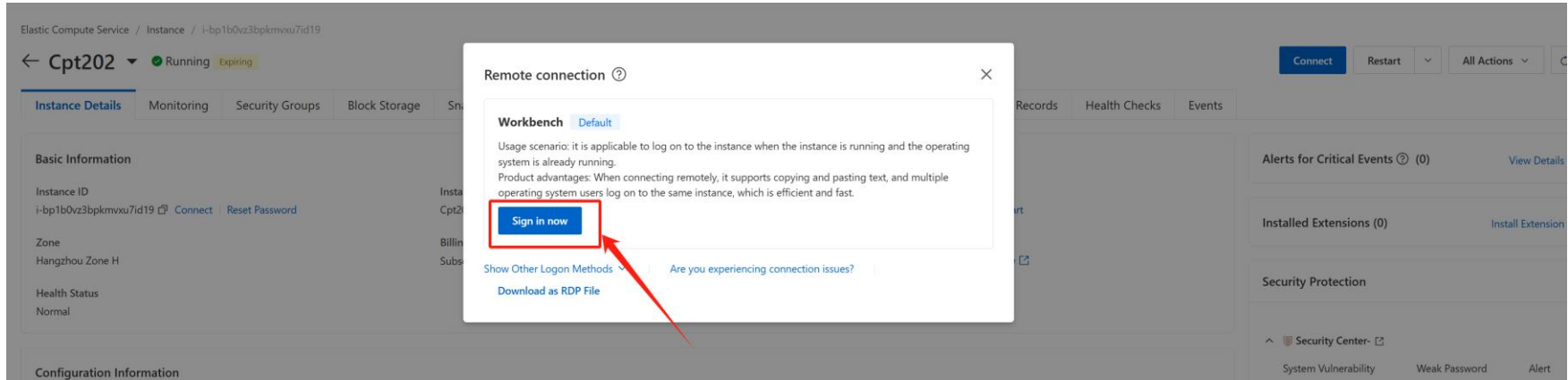
Instance Type ecs.e-c1m2.large Change Purchase Same Specifications

CPU and Memory 4 Cores (vCPU) 8 GiB Change CPU Options Change Configurations

Public IP Address 47.96.158.212 Convert to EIP

Operating System Windows Server 2025 DataCenter Edition 64bit English Edition Change Image

ECS instance

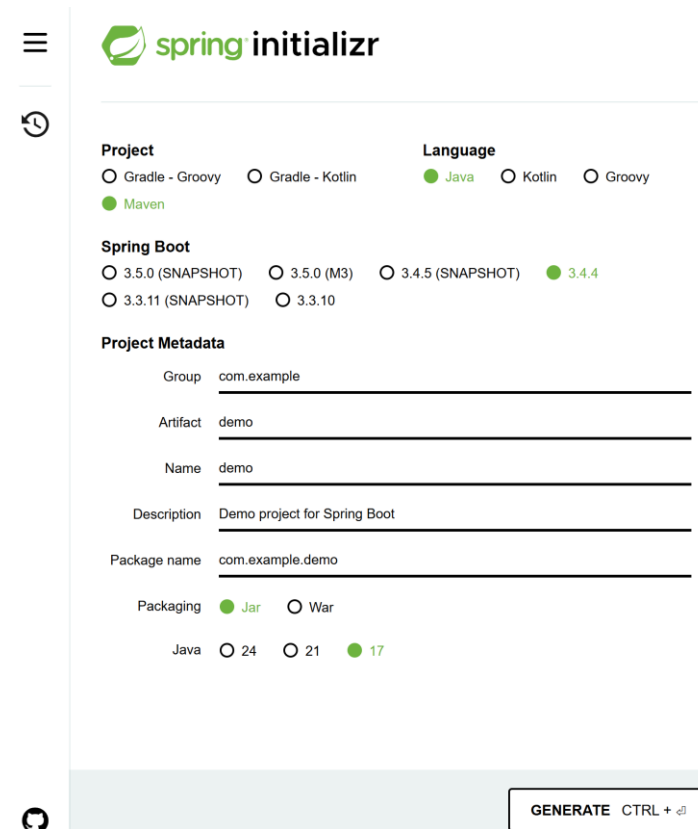


Prepare your Java project

1. Starting with Spring Initializer

- 1) Navigate to <https://start.spring.io>. This service pulls in all the dependencies you need for an application and does most of the setup for you.
- 2) Choose either Maven (in this module) and the language (Java) you want to use.
- 3) Click Dependencies and select Spring Web.
- 4) Click Generate.
- 5) Download the resulting ZIP file, which is an archive of a web application that is configured with your choices.

(If your IDE has the Spring Initializr integration, you can complete this process from your IDE)



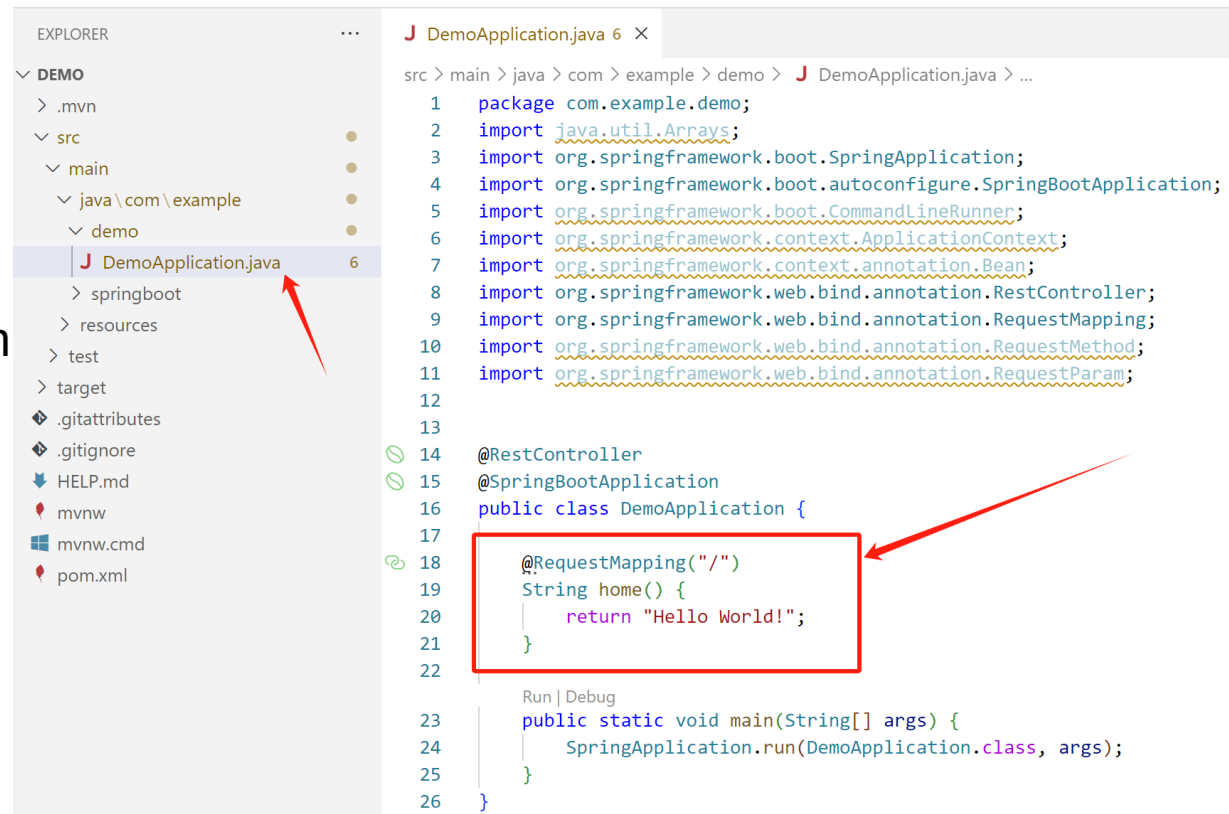
The screenshot shows the Spring Initializr web interface. It features a sidebar with a hamburger menu and a clock icon. The main form is titled 'spring initializr' and contains several sections: 'Project' with radio buttons for 'Gradle - Groovy', 'Gradle - Kotlin', and 'Maven' (selected); 'Language' with radio buttons for 'Java' (selected), 'Kotlin', and 'Groovy'; 'Spring Boot' with radio buttons for '3.5.0 (SNAPSHOT)', '3.5.0 (M3)', '3.4.5 (SNAPSHOT)', '3.4.4' (selected), and '3.3.11 (SNAPSHOT)'; 'Project Metadata' with input fields for 'Group' (com.example), 'Artifact' (demo), 'Name' (demo), 'Description' (Demo project for Spring Boot), and 'Package name' (com.example.demo); and 'Packaging' with radio buttons for 'Jar' (selected) and 'War'. At the bottom right, there is a 'GENERATE CTRL + ⌘' button.

Prepare your Java project

Run the application:

Run the following command in a terminal window (in the complete) directory:

```
./mvnw spring-boot:run
```



```
EXPLORER
...
DEMO
  > .mvn
  > src
    > main
      > java\com\example
        > demo
          DemoApplication.java 6
          > springboot
          > resources
          > test
          > target
          .gitattributes
          .gitignore
          HELP.md
          mvnw
          mvnw.cmd
          pom.xml

J DemoApplication.java 6 X
src > main > java > com > example > demo > J DemoApplication.java > ...
1  package com.example.demo;
2  import java.util.Arrays;
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5  import org.springframework.boot.CommandLineRunner;
6  import org.springframework.context.ApplicationContext;
7  import org.springframework.context.annotation.Bean;
8  import org.springframework.web.bind.annotation.RestController;
9  import org.springframework.web.bind.annotation.RequestMapping;
10 import org.springframework.web.bind.annotation.RequestMethod;
11 import org.springframework.web.bind.annotation.RequestParam;
12
13
14 @RestController
15 @SpringBootApplication
16 public class DemoApplication {
17
18     @RequestMapping("/")
19     String home() {
20         return "Hello World!";
21     }
22
23     Run | Debug
24     public static void main(String[] args) {
25         SpringApplication.run(DemoApplication.class, args);
26     }
27 }
```


Package your Java project

1. Compile Project on local computer:

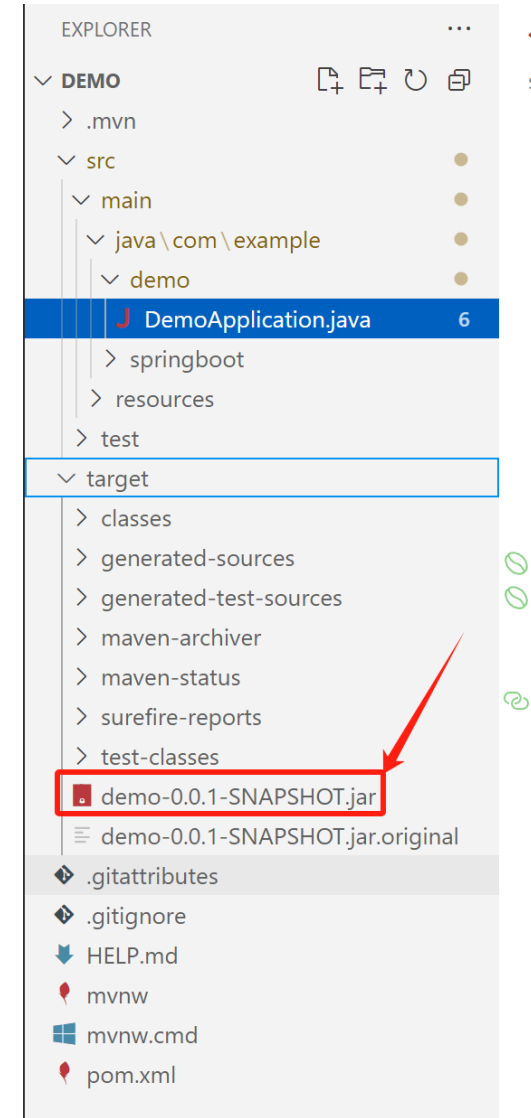
`./mvnw spring-boot:run`

2. Package the Spring Boot Project:

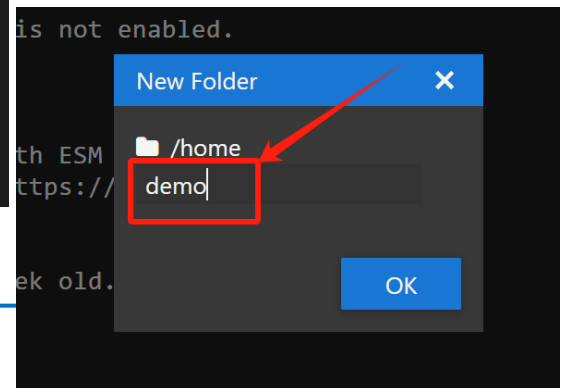
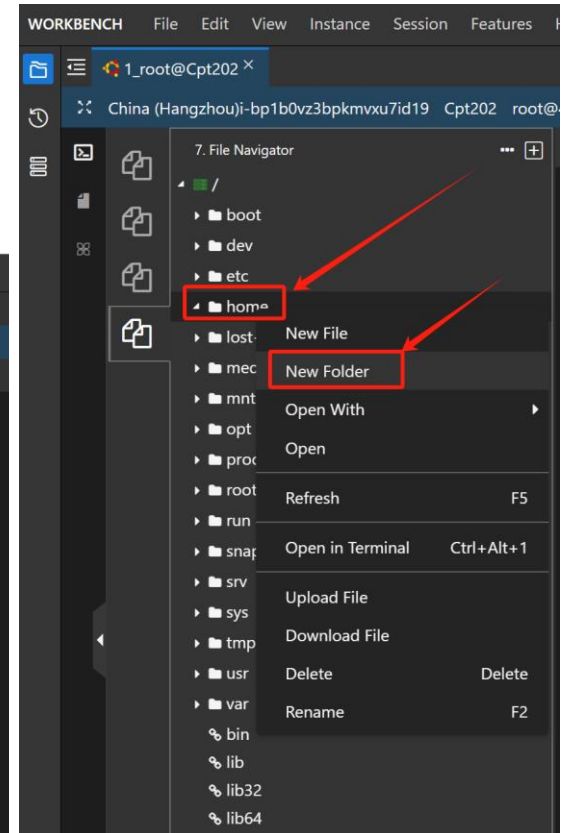
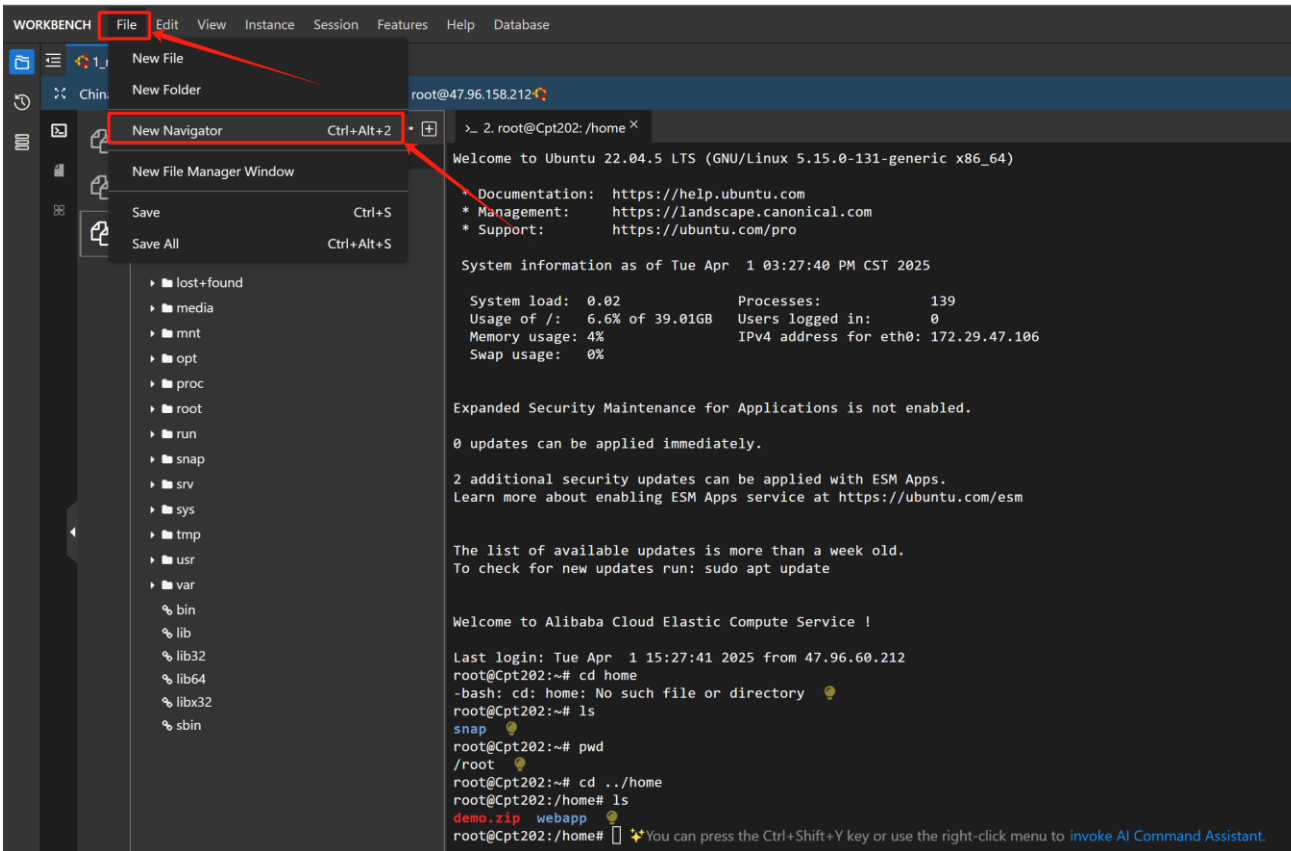
- In your local development environment, package the project using Maven:

`./mvnw package`

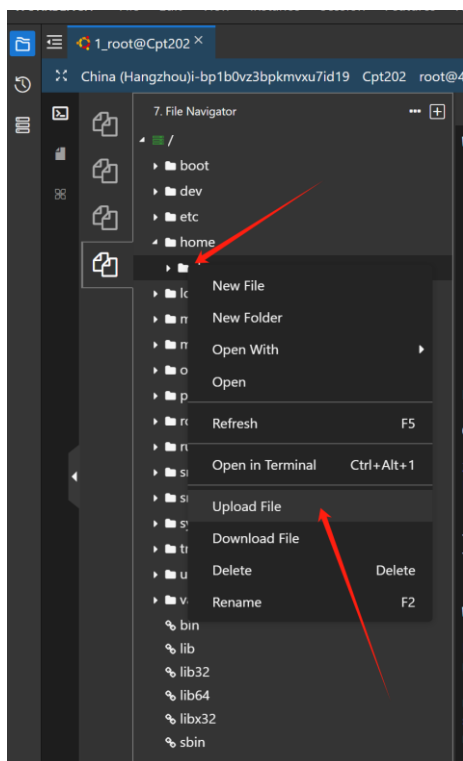
Locate the generated JAR file (usually in the `target` folder).



Upload your jar to server



Server Configuration



```
> 2. root@Cpt202: /home/demo X
root@Cpt202:~# pwd
/root
root@Cpt202:~# cd ..
root@Cpt202:~# ls
bin boot dev etc home lib lib32 lib64 libx32 lost+found media mnt opt proc root run sbin snap srv sys usr var
root@Cpt202:~# cd home
root@Cpt202:/home# ls
demo
root@Cpt202:/home# cd demo
root@Cpt202:/home/demo# ls
demo.zip
root@Cpt202:/home/demo# unzip -n demo.zip
Command 'unzip' not found, but can be installed with:
apt install unzip
root@Cpt202:/home/demo# apt install unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package unzip is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'unzip' has no installation candidate
root@Cpt202:/home/demo# apt install unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package unzip is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'unzip' has no installation candidate
root@Cpt202:/home/demo# sudo apt update
Get:1 http://mirrors.cloud.aliyuncs.com/ubuntu jammy InRelease [270 kB]
Get:2 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://mirrors.cloud.aliyuncs.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/main amd64 Packages [1,395 kB]
Get:6 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/main Translation-en [510 kB]
Get:7 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/main amd64 c-n-f Metadata [30.3 kB]
Get:8 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/restricted amd64 Packages [129 kB]
Get:9 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/restricted Translation-en [18.6 kB]
Get:10 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/restricted amd64 c-n-f Metadata [488 B]
Get:11 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:12 http://mirrors.cloud.aliyuncs.com/ubuntu jammy/universe Translation-en [5,652 kB]
```

File Tasks

Upload Tasks	Download Tasks
Upload All Failed Items Again	Cancel All
demo.zip	20.6 K
demo.zip	20.6 K

```
root@Cpt202:/home/demo/demo# ls
HELP.md  mvnw  mvnw.cmd  pom.xml  src  target
root@Cpt202:/home/demo/demo# java -v
Command 'java' not found, but can be installed with:
apt install openjdk-11-jre-headless # version 11.0.26+4-1ubuntu1~22.04, or
apt install default-jre             # version 2:1.11-72build2
apt install openjdk-17-jre-headless # version 17.0.14+7-1~22.04.1
apt install openjdk-18-jre-headless # version 18.0.2+9-2~22.04
apt install openjdk-19-jre-headless # version 19.0.2+7-0ubuntu3~22.04
apt install openjdk-21-jre-headless # version 21.0.6+7-1~22.04.1
apt install openjdk-8-jre-headless  # version 8u442-b06~us1-0ubuntu1~22.04
root@Cpt202:/home/demo/demo# apt install openjdk-17-jre-headless
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2
Suggested packages:
  default-jre libasound2-plugins alsa-utils liblcms2-utils pcsd libnss-mdns fonts-indic
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java java-common libasound2
0 upgraded, 11 newly installed, 0 to remove and 56 not upgraded.
Need to get 49.3 MB of archives.
After this operation, 197 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

(Navigate to the folder and) Run the JAR File

Execute the JAR file:

```
java -jar app.jar
```

- If the app uses a specific port (e.g., 8080), ensure the port is open in the firewall:

```
sudo ufw allow 8080
```

Access the Web App

Open a browser and navigate to:

<http://your-server-public> ip:port

Replace `port` with the port your app uses (e.g., 8080).

MySQL Server Configuration

1. Update System Packages

Before installing MySQL, it is recommended to update the system package list:

```
sudo apt update
```

2. Install MySQL

Install MySQL using the following command:

```
sudo apt install mysql-server
```

During the installation, you may be prompted to set a password for the MySQL root user. Follow the prompts to complete the setup.

3. Configure MySQL

After installation, run the following command to configure MySQL security options:

```
sudo mysql_secure_installation
```

This script will guide you through the following configurations:

- Set a password for the root user (if not already set).
- Remove anonymous users.
- Disable remote login for the root user (by default, root can only log in locally).
- Delete test databases.
- Reload the privilege tables.

```
root@cpt202:~# sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  bolt cups-bsd cups-client debugedit dh-autoreconf dh-strip-nondeterminism dwz fwupd-signed
  glib1.2-packagekit-glib1.0 intel-microcode lucode-tool libarchive-cpio-perl libarchive-zip-perl libatasmart4
  libblockdev-cryptot2 libblockdev-fs2 libblockdev-loop2 libblockdev-part-err2 libblockdev-part2 libblockdev-swap2
  libblockdev-util2 libblockdev2 libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libdebhelper-perl
  libfile-stripnondeterminism-perl libfsverity0 libfwupd2 libglib-1.0-0 libgpgme11 libgstreamer1.0-0
  libgudev-1.0-0 libgusb2 libinih1 libintl-perl libintl-xs-perl libjcat1 libjson-glib-1.0-0
  libjson-glib-1.0-common libldb0 libmail-sendmail-perl libmariadb3 libmaxminddb0 libmbim-glib4 libmbim-proxy
  libmm-glib0 libmodule-find-perl libmodule-scandeps-perl libmspack0 libpackagekit-glib2-18 libparted-fs-resize0
  libproc-processtable-perl libqmi-glib5 libqmi-proxy librpm9 librpm-build9 librpmio9 librpm-sign9 libsbios-c2
  libsort-naturally-perl libstemmer0 libsub-override-perl libsys-hostname-long-perl libterm-readkey-perl
  libudisks2-0 libvolume-key1 libxmlb2 libx-security mariadb-common modemmanager pax python3-software-properties
  rpm-common udisks2 usb-modeswitch usb-modeswitch-data zerofree
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libfcgi-fast-perl libfcgi-pm-perl libclone-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libbusiness-isbn-perl libwww-perl tinyca
The following packages will be REMOVED:
  mariadb-client mariadb-client-10.6 mariadb-client-core-10.6
The following NEW packages will be installed:
  libfcgi-fast-perl libfcgi-pm-perl libclone-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-server mysql-server-8.0
mysql-server-core-8.0
0 upgraded, 26 newly installed, 3 to remove and 27 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 202 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
root@cpt202:~# sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW      Length >= 8
MEDIUM  Length >= 8, numeric, mixed case, and special characters
STRONG  Length >= 8, numeric, mixed case, special characters and dictionary      file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: 0

Skipping password set for root as authentication with auth_socket is used by default.
If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management for more information.

By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : n

... skipping.
By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : n

... skipping.
Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
```


MySQL Server Configuration

4. Start and Enable MySQL Service

Ensure the MySQL service is running and set to start on boot:

```
sudo systemctl start mysql  
sudo systemctl enable mysql
```

```
root@Cpt202:~# sudo systemctl start mysql
```

```
root@Cpt202:~# sudo systemctl enable mysql
```

Synchronizing state of mysql.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mysql

5. Configure Remote Access

5.1 Modify MySQL Configuration File

Edit the MySQL configuration file to allow remote connections:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

Find the bind-address line and change 127.0.0.1 to 0.0.0.0:

```
bind-address = 0.0.0.0
```

Save and exit (press Ctrl+O to save, Ctrl+X to exit).

5.2 Restart MySQL Service

```
sudo systemctl restart mysql
```

5.3 Configure Firewall

Ensure the firewall allows traffic on MySQL's default port (3306):

```
sudo ufw allow 3306
```

```
>_ 3. root@Cpt202: ~ X  
root@Cpt202:~# sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf  
root@Cpt202:~# sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf  
root@Cpt202:~# sudo systemctl restart mysql  
root@Cpt202:~# sudo systemctl enable mysql  
Synchronizing state of mysql.service with SysV service script with /lib/systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable mysql  
root@Cpt202:~# sudo ufw allow 3306  
Rule added  
Rule added (v6)  
root@Cpt202:~#
```

```
>_ 3. root@Cpt202: ~ X  
GNU nano 6.2 /etc/mysql/mysql.conf.d/mysqld.cnf  
#  
# The MySQL database server configuration file.  
#  
# One can use all long options that the program supports.  
# Run program with --help to get a list of available options and with  
# --print-defaults to see which it would actually understand and use.  
#  
# For explanations see  
# http://dev.mysql.com/doc/mysql/en/server-system-variables.html  
#  
# Here is entries for some specific programs  
# The following values assume you have at least 32M ram  
[mysqld]  
#  
# * Basic Settings  
#  
user = mysql  
# pid-file = /var/run/mysqld/mysqld.pid  
# socket = /var/run/mysqld/mysqld.sock  
# port = 3306  
# datadir = /var/lib/mysql  
#  
# If MySQL is running as a replication slave, this should be  
# changed. Ref https://dev.mysql.com/doc/refman/8.0/en/server-system-variables.html#sysvar_tmpdir  
# tmpdir = /tmp  
#  
# Instead of skip-networking the default is now to listen only on  
# localhost which is more compatible and is not less secure.  
bind-address = 127.0.0.1  
mysqlx-bind-address = 127.0.0.1  
#  
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo  
^X Exit ^R Read File ^\ Replace ^U Paste ^D Justify ^_ Go To Line M-E Redo
```

5.5 Test MySQL Connection

You can test whether MySQL is running correctly by running:

```
mysql -u root -p
```

If you are able to log in successfully after entering the password, the installation is complete.

5.6 Create a Remote Access User

Log into MySQL and create a user who can access the database remotely:

```
sudo mysql -u root -p
```

Once inside MySQL, run the following commands:

```
CREATE USER 'remote_user'@'%' IDENTIFIED BY 'your_password'; GRANT ALL PRIVILEGES ON *.* TO  
'remote_user'@'%' WITH GRANT OPTION; FLUSH PRIVILEGES;
```

Replace `remote_user` and `your_password` with your desired username and password.

Common Issues

1.Forgot the Root Password: You can reset the root password using the following commands:

```
sudo mysqld_safe --skip-grant-tables & sudo mysql -u root mysql UPDATE user SET  
Password=PASSWORD('new_password') WHERE User='root'; FLUSH PRIVILEGES;
```

2.Remote Connection Failure: Check firewall rules, security group configurations, and the MySQL `bind-address` setting.

```
>_ 3.root@Cpt202: ~ ×  
root@Cpt202:~# sudo mysql -u root -p  
Enter password:  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 8  
Server version: 8.0.41-0ubuntu0.22.04.1 (Ubuntu)  
  
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Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
mysql>
```




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WebApp Deployment

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